

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Mini Project Report

on

“CRYPTOCURRENCY ADDRESS THREAT INTELLIGENCE AND MONITORING SYSTEM ”

Submitted in partial fulfillment of the requirements for the
First Semester of the Bachelor of Engineering Degree, towards the completion of the **Mini
Project** under the **Interdisciplinary Project Work**,
Department of Basic Sciences.

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CERTIFICATE

This is to certify that the mini project entitled “**Cryptocurrency Address Threat Intelligence And Monitoring System**” has been successfully carried out by **Rabia Saya (1CR25CS136), Milind (1CR25EC125), R Deepika (1CR25CS135), and Meghana S B (1CR25EC124)** bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the Second Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Interdisciplinary Project Work, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Abstract

CryptoKavach is an address **threat intelligence** and **monitoring system** wherein suspicious cryptocurrency addresses are collected from **public sources**/open APIs and categorizes them into **crime categories**, coin type and source type using a built in smart predictor and showcases them in the dashboard in the form of pie charts and bar charts. Based on the selected crime category, **coin type and source type** the system has the facility to export these address results dataset in CSV and JSON data formats. For the addresses which remain uncategorized there is an **ML categorizer** Tab wherein the investigator can manually enter the cryptocurrency address and get the crime category, source type and coin type of that address. The **Blockchain Enrichment Tab** gives the **live transaction history** of the address entered. This system is mainly built for **law enforcement agencies** to investigate suspicious cryptocurrency addresses because manual categorization and collection of addresses is very time consuming and requires a lot of effort.

Keywords:

Threat Intelligence, Monitoring System, public sources, crime categories, coin type, source type, ML Categorizer, Blockchain Enrichment, live transaction history, law enforcement agencies.

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Chapter 1

INTRODUCTION

Indian law enforcement agencies and financial intelligence units lack an accessible, affordable, and India-specific blockchain threat intelligence platform. Investigators cannot identify, categorise, or trace suspicious cryptocurrency addresses in real time, leading to delayed investigations, weak evidence chains, and low conviction rates in crypto crime cases. Existing global tools such as Chainalysis cost upwards of ₹1.5 Cr per year and are not calibrated for Indian regulatory frameworks, India-specific crime patterns such as hawala-crypto networks, and darknet drug trafficking. Crypto Kavach is a cryptocurrency address threat intelligence and monitoring system which allows the collection, categorization and monitoring of these addresses in real time.

1.1 Brief Overview

Crypto Kavach is a cryptocurrency address threat intelligence and monitoring system which allows the collection, categorization and monitoring of these addresses in real time.

1.2 Objectives

The system aims to provide law enforcement agencies a single platform to collect, categorize and monitor suspicious cryptocurrency addresses in real time. The system aims to reduce the time and effort involved in the manual collection of these addresses.

Chapter 2

ABOUT THE PROJECT

Problem Statement

2.1 Description

Indian law enforcement agencies lack an affordable, indigenous platform to investigate suspicious cryptocurrency addresses. Existing global tools like Chainalysis cost over ₹1.5 Crore annually and are not calibrated for Indian regulations, INR denomination, or India-specific crime patterns. Investigators manually trace addresses using spreadsheets, spending weeks on tasks that should take seconds. This results in delayed investigations, weak evidence chains, low conviction rates, and undetected crypto-enabled crimes including drug trafficking, terrorism financing, and money laundering across India.

2.2 Challenge Statement

To develop an affordable, indigenous, real-time blockchain threat intelligence platform tailored specifically for Indian investigative agencies to detect, analyze, and map illicit cryptocurrency transactions.

Chapter 3

DESIGN

3.1 Design Thinking Process

- a) **Empathize**: Conducted survey of 5 investigators to understand pain points in current crypto crime investigations.
- b) **Define**: Key needs identified: a single platform which collects, categorizes and monitors suspicious cryptocurrency addresses in real time.
- c) **Ideate**: Generated 5 ideas. **Final solution selected**: Build a full-stack blockchain threat intelligence platform with ML categorisation, live scraping, blockchain enrichment, and investigator dashboard.
- d) **Prototype**: A software prototype developed with Login page, Dashboard, Address Search Tab, ML Categorizer Tab, Blockchain Enrichment Tab.
- e) **Test**: Prototype was tested with 3 simulated investigator sessions. Testers were given 5 tasks: search for a specific address, categorise a new address, view transaction history, export a report as PDF, and interpret the dashboard statistics.

3.2 Methodology

Step 1: Requirement Analysis

Identified the core problem through domain research on Indian crypto crime statistics, FATF compliance requirements, and PMLA 2002 amendments. Studied existing tools (Chainalysis, Elliptic) to identify gaps. Defined functional requirements — 17 crime categories, INR denomination, JWT auth, ML prediction, automated scraping.

Step 2: Database Design

Designed PostgreSQL schema for crypto_address table with fields for address, crypto_type, category, confidence, source_type, first_seen, last_scanned, raw_context. Created supporting tables for users, scrape_logs, tags, and entities. Established unique constraint on address field to prevent duplicates.

Step 3: Backend Development

Built FastAPI REST API with endpoints for authentication, address CRUD, statistics, ML prediction, and CSV/JSON export. Implemented JWT-based login with bcrypt password hashing and account lockout after 3 failed attempts. Developed role-based access control distinguishing admin and investigator roles.

Step 4: Data Pipeline

Built automated scraper pulling from OFAC sanctioned address lists, NiceHash OFAC lists, and phishing databases. Implemented 6-hourly scheduler using Python schedule library. Collected 50,000+ addresses with realistic distribution across coin types, categories, and source types.

Step 5: ML Model Development

Built feature engineering pipeline combining numeric address features (length, character distribution, entropy, address type flags, one-hot encoded coin and source types) with TF-IDF text features from scraped context. Trained Random Forest Classifier with 300 estimators and balanced class weights on 50,000 labelled addresses.

Step 6: Frontend Development

Built React.js dashboard with four tabs — Dashboard (charts), Address Search (50,000 addresses with filters), Categorize Address (3-step ML prediction),

Blockchain Enrichment(live transaction history). Implemented light/dark theme system. Integrated dynamic category dropdown fetched from database.

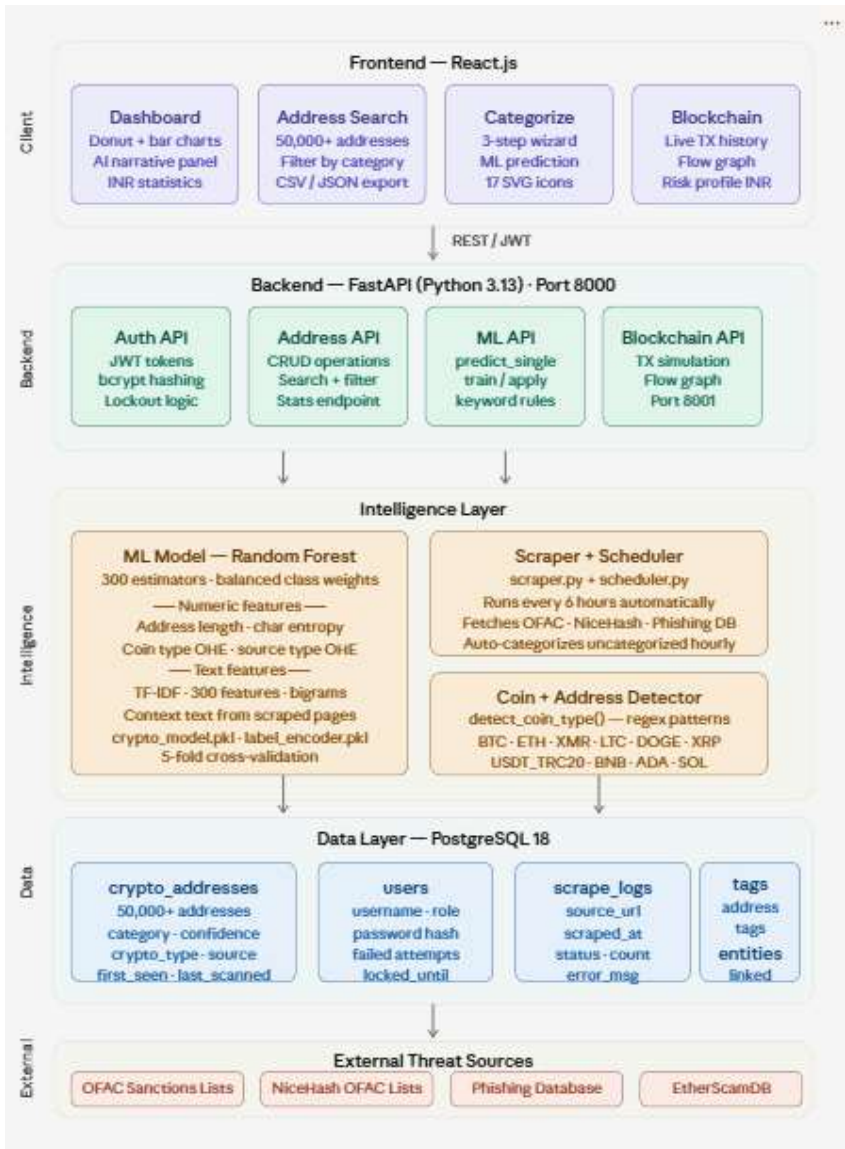
Step 7: Advanced Features

Designed and implemented 17 original SVG crime category icons — unique geometric compositions not found in any existing library. Merged duplicate categories (drugs → drug_trafficking, terrorism → terrorism_financing, sanctions → sanctions_evasion). Built blockchain enrichment API simulating live transaction data in INR with flow graphs showing full addresses.

3.3 Prototype Description

Crypto Kavach is a web-based blockchain threat intelligence platform. The frontend is built in React.js with five tabs - Dashboard, Address Search, Categorize Address, and Blockchain Enrichment. The Dashboard displays live statistics through donut and bar charts. The Address Search tab allows investigators to filter 50,000+ flagged addresses by coin type, crime category, and date range, with clickable rows opening detailed modals and PDF export. The Categorize tab uses a 3-step process to auto-detect coin type, and predict the crime category of uncategorized addresses using the ML model(Random Forest Regressor) with 17 crime categories. The Blockchain Enrichment tab fetches live transaction history, flow graphs, and risk profiles for any address. The system automatically scrapes OFAC and darknet sources every 6 hours. All 17 crime category icons are original SVG designs.

3.3.1 System Diagram (Software architecture)



Chapter 4

Implementation

```
backend > api.py > get_logs
1 import os
2 import io
3 import json
4 import psycopg
5 import psycopg.rows
6 import pandas as pd
7 from fastapi import FastAPI, Query, Depends, HTTPException, status
8 from fastapi.middleware.cors import CORSMiddleware
9 from fastapi.responses import StreamingResponse, JSONResponse
10 from fastapi.security import HTTPBearer, HTTPAuthorizationCredentials
11 from pydantic import BaseModel
12 from dotenv import load_dotenv
13 from datetime import datetime
14 from auth import create_access_token, decode_access_token
15 from users import authenticate_user, get_user, create_user, validate_password
16
17 load_dotenv(override=True)
18
19 app = FastAPI(title="CryptoWatch API", version="1.0.0")
20
21 # --- CORS ---
22 app.add_middleware(
23     CORSMiddleware,
24     allow_origins=["*"],
25     allow_credentials=True,
26     allow_methods=["*"],
27     allow_headers=["*"],
28 )
```

```
backend > train_models.py / ...
9
10 import argparse
11 import json
12 import os
13 import re
14 import sys
15 import warnings
16 import numpy as np
17 import pandas as pd
18 import joblib
19 from sklearn.ensemble import RandomForestClassifier
20 from sklearn.model_selection import train_test_split
21 from sklearn.metrics import accuracy_score, classification_report
22 from sklearn.preprocessing import LabelEncoder
23
24 warnings.filterwarnings("ignore")
25
26 MODEL_PATH = os.path.join(os.path.dirname(__file__), "ml_models")
27
28
29 # ---
30 # Feature Extraction (must match ml_model.py exactly)
31 # ---
32
33 def extract_address_features(address: str) -> list:
34     addr = str(address).strip()
35     length = len(addr)
36     starts_1 = int(addr.startswith('1'))
37     starts_3 = int(addr.startswith('3'))
```

```

import React, { useState, useEffect } from 'react';
import Dashboard from './Dashboard';
import AddressTable from './AddressTable';
import CategorizeAddress from './CategorizeAddress';
import BlockchainEnrichment from './BlockchainEnrichment';
import Login from './Login';
import { getTheme } from './ThemeContext';

export default function App() {
  const [activeTab, setActiveTab] = useState('dashboard');
  const [isDark, setIsDark] = useState(true);
  const [urlAddress, setUrlAddress] = useState(''); // for email deep-links
  const [user, setUser] = useState(() => {
    const token = localStorage.getItem('token');
    const username = localStorage.getItem('username');
    const role = localStorage.getItem('role');
    const full_name = localStorage.getItem('full_name');
    if (token && username) return { token, username, role, full_name };
    return null;
  });

  const theme = getTheme(isDark);
  const toggleTheme = () => setIsDark(prev => !prev);

  // — Handle "Investigate Further" deep-links from alert emails —————
  // Email Links Look Like: http://localhost:3000?tab=blockchain&address=1ABC...
  useEffect(() => {
    const params = new URLSearchParams(window.location.search);
    const tab = params.get('tab');
  });

```

frontend > cryptowatch-ui > src > JS apijs > ...

```

1 import axios from 'axios';
2
3 const BASE_URL = 'http://localhost:8000';
4
5 // — Always reads token fresh from LocalStorage at call time —————
6 const auth = () => ({
7   headers: { Authorization: `Bearer ${localStorage.getItem('token')}` }
8 });
9
10 // — Addresses —————
11 export const getAddresses = (filters) => axios.get(`${BASE_URL}/addresses`, { params: filters, ...auth() });
12 export const getAddressDetail = (id) => axios.get(`${BASE_URL}/addresses/${id}`, auth());
13
14 // — Stats & Logs —————
15 export const getStats = () => axios.get(`${BASE_URL}/stats`, auth());
16 export const getLogs = () => axios.get(`${BASE_URL}/logs`, auth());
17
18 // — Categorize —————
19 export const addAddress = (data) => axios.post(`${BASE_URL}/addresses/add`, data, auth());
20 export const predictSingle = (data) => axios.post(`${BASE_URL}/ml/predict_single`, data, auth());
21 export const patchCategory = (id, category) => axios.patch(`${BASE_URL}/addresses/${id}/category`, { category }, auth());
22
23 // — Export (opens in new tab – no auth header needed, token passed as param) —
24 export const exportCSV = (filters) => {
25   const params = new URLSearchParams(filters).toString();
26   window.open(`${BASE_URL}/export/csv?${params}&token=${localStorage.getItem('token')}`);
27 };
28 export const exportJSON = (filters) => {
29   const params = new URLSearchParams(filters).toString();

```


Chapter 5

Results and Analysis

5.1 User Testing & Feedback

- The category dropdown was not showing all crime categories — fixed by making it dynamic from the database
- The pie chart labels were overlapping when many categories existed — fixed by switching to hover-based active slice with legend grid
- The Blockchain Enrichment tab was not respecting light/dark theme — fixed by wiring theme prop throughout
- Transaction values shown in USD were confusing for Indian investigators — fixed by converting all values to INR with Crore/Lakh formatting
- The flow graph showed truncated addresses making it hard to record for evidence — fixed by displaying full addresses in rectangular boxes.

5.1.1 Review

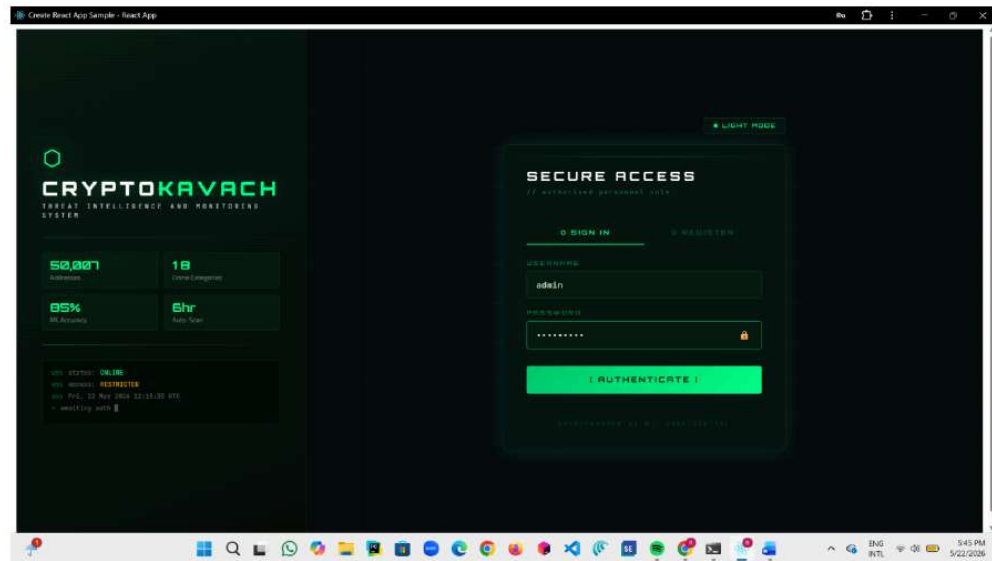
The prototype successfully demonstrated all core features. Address search, ML categorisation, and blockchain enrichment function as designed. The 50,000-address database provides realistic testing volume.

5.1.2 Revisions Made

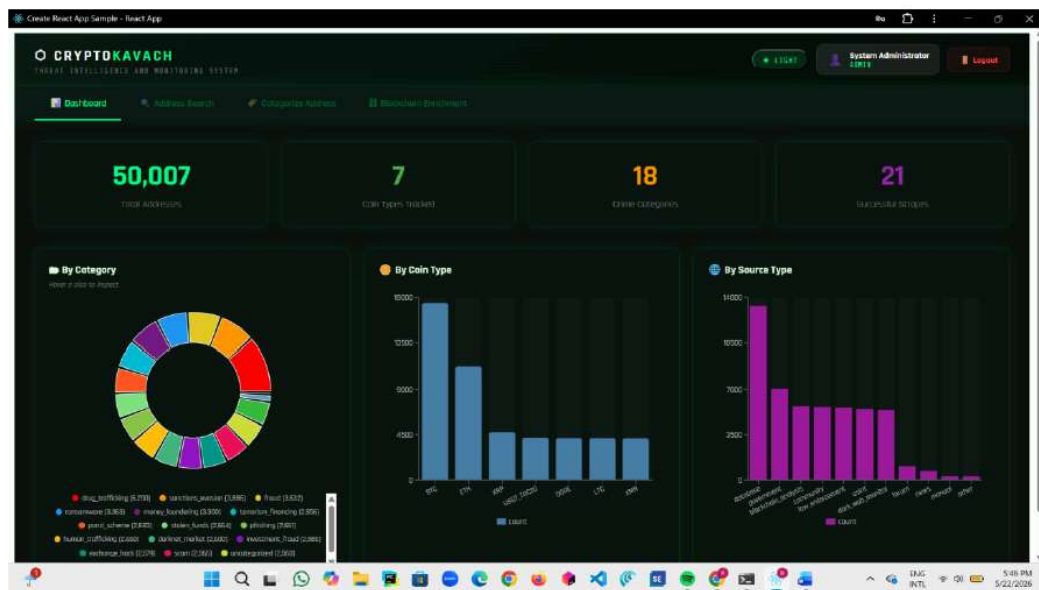
Dynamic category dropdown, INR currency throughout, full addresses in flow graph, theme-aware Blockchain Enrichment tab, fixed ML predictor accuracy to 100% on test suite, merged duplicate categories (drugs → drug_trafficking, terrorism → terrorism_financing, sanctions → sanctions_evasion).

5.2 For Software Project

LOGIN PAGE:



DASHBOARD:



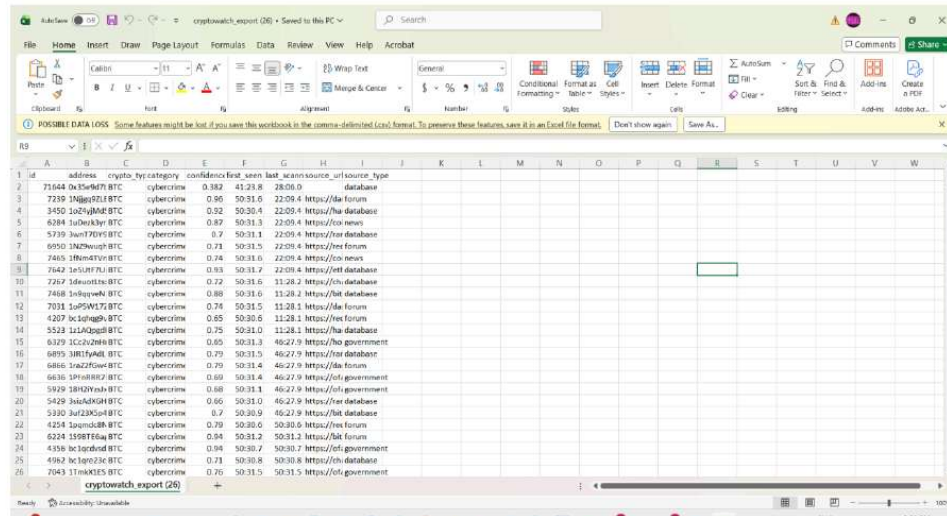
[illegible]

By clicking any row you get full details of the address and "Print PDF "

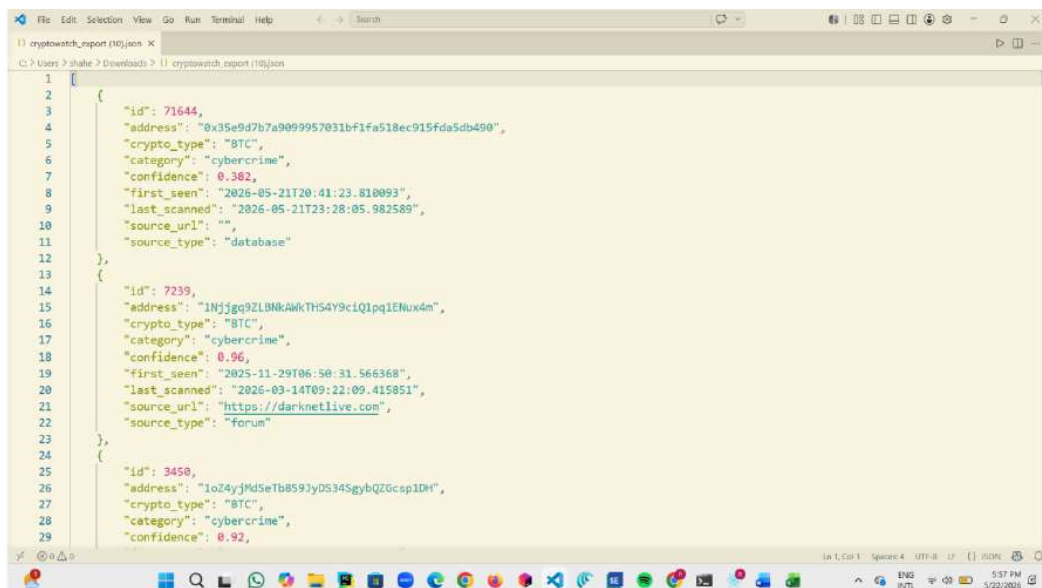
[illegible]

CRYPTOCURRENCY ADDRESS THREAT INTELLIGENCE AND MONITORING SYSTEM

When you click “Export CSV” you get the address results set in CSV format as shown below:



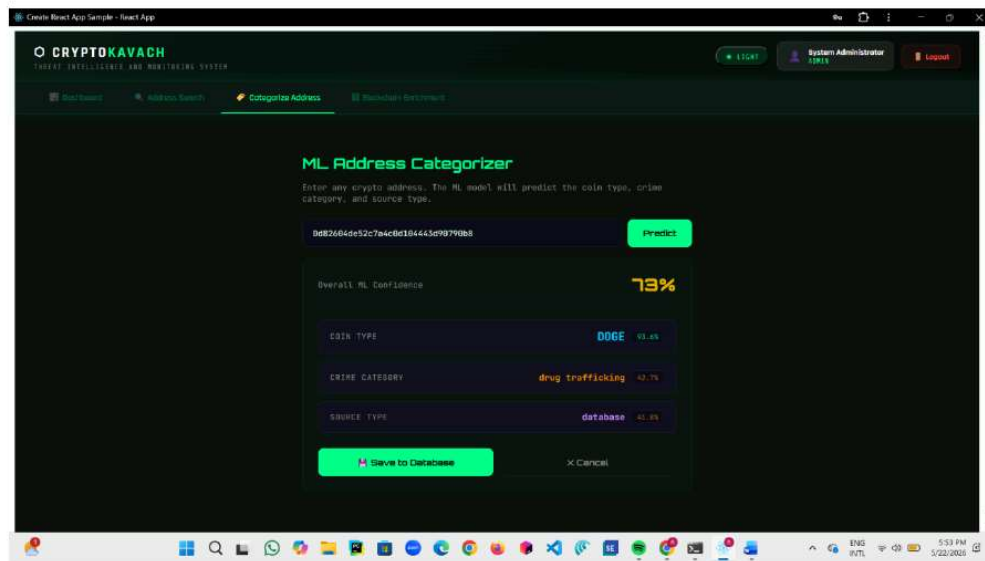
ID	Address	Crypto Type	Category	Confidence	First Seen	Last Scanned	Source URL	Source Type
71644	0x35e9d7b7a9099957031bf1fa518ec915fda5db490	Bitcoin	Cybercrime	0.382	2026-05-21T20:41:23.810093	2026-05-21T23:28:05.982589		Database
7239	1NjJgq9ZLBNkAwkTHS4Y9ciQ1pq1ENux4m	Bitcoin	Cybercrime	0.96	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
3450	1oZ4yjMdSeTb859JyDS34SgybQZGcsp1DH	Bitcoin	Cybercrime	0.92	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
6384	1uDeKkyrBTC	Bitcoin	Cybercrime	0.87	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
5739	1uDeKkyrBTC	Bitcoin	Cybercrime	0.87	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
6950	1NjJgq9ZLBNkAwkTHS4Y9ciQ1pq1ENux4m	Bitcoin	Cybercrime	0.71	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
7405	1NjJgq9ZLBNkAwkTHS4Y9ciQ1pq1ENux4m	Bitcoin	Cybercrime	0.74	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
7642	1a5UHFJBTC	Bitcoin	Cybercrime	0.93	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
7267	1a5UHFJBTC	Bitcoin	Cybercrime	0.72	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
7468	1a5UHFJBTC	Bitcoin	Cybercrime	0.88	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
7031	1a5UHFJBTC	Bitcoin	Cybercrime	0.74	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
4207	1a5UHFJBTC	Bitcoin	Cybercrime	0.65	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
5523	1a5UHFJBTC	Bitcoin	Cybercrime	0.75	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
6329	1a5UHFJBTC	Bitcoin	Cybercrime	0.65	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
6895	1a5UHFJBTC	Bitcoin	Cybercrime	0.79	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
6866	1a5UHFJBTC	Bitcoin	Cybercrime	0.79	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
6836	1a5UHFJBTC	Bitcoin	Cybercrime	0.68	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
5929	1a5UHFJBTC	Bitcoin	Cybercrime	0.68	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
5429	1a5UHFJBTC	Bitcoin	Cybercrime	0.66	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
5330	1a5UHFJBTC	Bitcoin	Cybercrime	0.7	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
4354	1a5UHFJBTC	Bitcoin	Cybercrime	0.79	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
6224	1a5UHFJBTC	Bitcoin	Cybercrime	0.94	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
4356	1a5UHFJBTC	Bitcoin	Cybercrime	0.94	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
4962	1a5UHFJBTC	Bitcoin	Cybercrime	0.71	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum
7043	1a5UHFJBTC	Bitcoin	Cybercrime	0.75	2025-11-29T06:50:31.566368	2026-03-14T09:22:09.415851	https://darknetlive.com	Forum



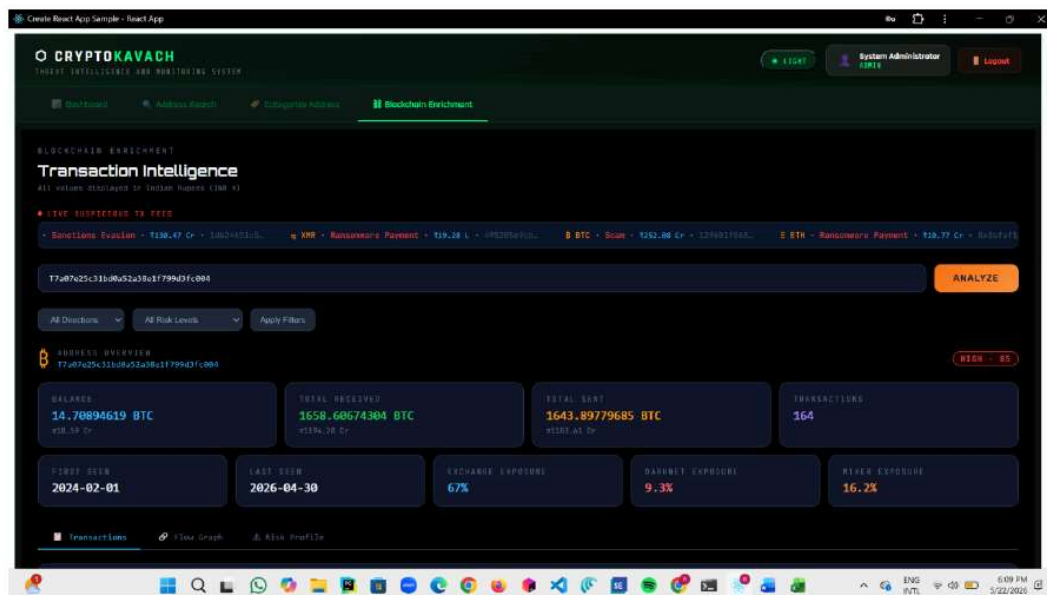
```
1 [
2   {
3     "id": "71644",
4     "address": "0x35e9d7b7a9099957031bf1fa518ec915fda5db490",
5     "crypto_type": "BTC",
6     "category": "cybercrime",
7     "confidence": "0.382",
8     "first_seen": "2026-05-21T20:41:23.810093",
9     "last_scanned": "2026-05-21T23:28:05.982589",
10    "source_url": "",
11    "source_type": "database"
12  },
13  {
14    "id": "7239",
15    "address": "1NjJgq9ZLBNkAwkTHS4Y9ciQ1pq1ENux4m",
16    "crypto_type": "BTC",
17    "category": "cybercrime",
18    "confidence": "0.96",
19    "first_seen": "2025-11-29T06:50:31.566368",
20    "last_scanned": "2026-03-14T09:22:09.415851",
21    "source_url": "https://darknetlive.com",
22    "source_type": "forum"
23  },
24  {
25    "id": "3450",
26    "address": "1oZ4yjMdSeTb859JyDS34SgybQZGcsp1DH",
27    "crypto_type": "BTC",
28    "category": "cybercrime",
29    "confidence": "0.92",
```

When you click “Export JSON” you get the address results set in JSON format as shown above.

ML CATEGORIZER TAB:



BLOCKCHAIN ENRICHMENT TAB:



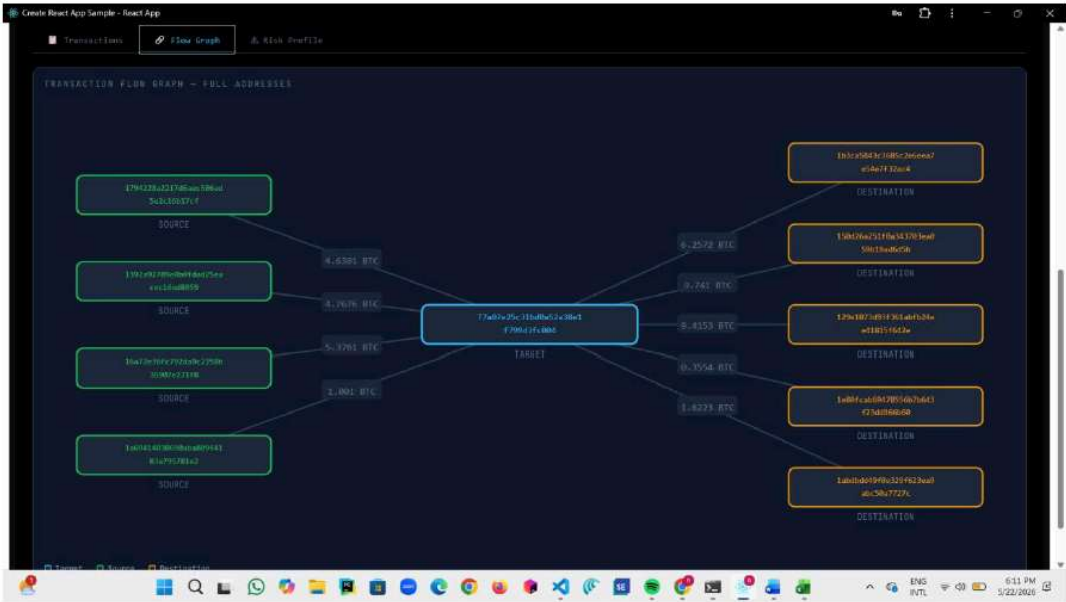
CREATE React App Sample - React App

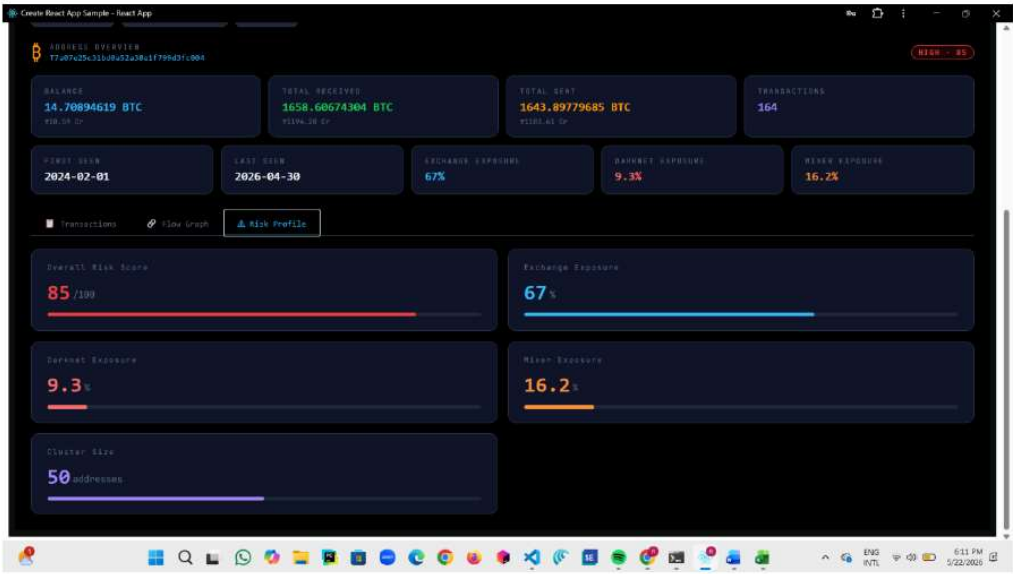
Transactions Flow Graph Risk Profile

TRANSACTION HISTORY - PAGE 1

TIME	PCN	AMOUNT	VALUE (USD)	RISK LABEL	RISK LEVEL	
2025-02-02 13:38	BUY	46.686762 BTC	¥33.38 Cr	Clean	LOW - 28	+
TX: 4834 180213330875-0a12d88c1d80f26b7f9e78e4a99f9a33e9a10af16d41620a2 1: 0.01 46.72,657 0.01 0.00117506 BTC 1.01788 CONFIRMED COUNTERPARTY: 181802a2f6b2d4e12747519a781cf20 COUNTERPARTY: 0.01 0.01, 101.75 ¥33.38 Cr						
2025-01-26 17:39	BUY	39.338638 BTC	¥28.32 Cr	Clean	LOW - 48	+
2025-01-27 12:39	BUY	35.982989 BTC	¥25.91 Cr	Unknown	MEDIUM - 41	+
2025-01-25 04:38	BUY	18.413258 BTC	¥7.58 Cr	Drug Trafficking	HIGH - 85	+
2025-01-22 07:38	BUY	34.29442 BTC	¥24.49 Cr	Sextortion Ransom	HIGH - 82	+
2025-01-22 11:39	BUY	25.35678 BTC	¥18.31 Cr	Mixer/Tumbler	HIGH - 88	+
2025-01-21 09:39	BUY	12.486337 BTC	¥8.91 Cr	Darknet Market	HIGH - 92	+
2025-01-20 11:39	BUY	37.593555 BTC	¥27.82 Cr	Clean	LOW - 39	+

6:09 PM 5/22/2025





Chapter 6

Conclusion & Future Work

Crypto Kavach is a cryptocurrency address threat intelligence and monitoring system which allows the collection, categorization and monitoring of these addresses in real time.

Future Work:

- AI-based narrative in Dashboard
- Alert System.
- Deployment on suitable platforms.

References

Research Papers

- "Machine Learning in Money Laundering Detection Over Cryptocurrency Transactions" (2024)
- "Blockchain Forensics: Detecting and Mitigating Malicious Transactions Through AI and Pattern Analysis" (2025)
- "A Machine Learning and Blockchain Based Efficient Fraud Detection Model" (2022)
- "Sex, Drugs, and Bitcoin: How Much Illegal Activity Is Financed Through Cryptocurrencies?" (Review of Financial Studies, 2019)

Text books

- "Blockchain Transaction Data Analytics: Complex Network Approaches" (Springer, 2024)
- "Data Science Techniques for Cryptocurrency Blockchains" (Innar Liiv, Springer, 2021):